JAWAD UL HADI

Backend Lead Engineer | Backend Architecture | AI-First Systems Design

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PROFESSIONAL SUMMARY

I'm a Backend Lead Engineer with 8+ years of experience building distributed systems for multi-tenant SaaS and enterprise products. At MicroAgility, I lead backend architecture for Talentnix, a multi-tenant SaaS platform, and the APAC Management System (HRMS). I work mainly in Node.js, NestJS, and TypeScript on PostgreSQL, MySQL, MongoDB, and Redis, with Python and FastAPI where they fit, and I own architecture from database schema to deployment. My recent focus is production AI: Anthropic-based features, MCP servers, provider-agnostic abstraction layers, and RAG pipelines, with retry and fallback patterns that keep failures quiet and contained. I also mentor three associate developers and work with stakeholders to turn requirements into agile delivery plans.

TECHNICAL SKILLS

Architecture & Systems Design: Distributed systems, system design, microservices, event-driven architecture, multi-tenant SaaS, REST and GraphQL API design, scalability and reliability trade-offs

Backend Engineering: Node.js, NestJS, TypeScript, Express.js, Python, FastAPI, Django REST Framework, MERN stack, PostgreSQL, MySQL, MongoDB, Redis, BullMQ, MeiliSearch, query and performance optimisation

AI-First Engineering: LLM integration (OpenAI, Gemini, Anthropic), RAG pipelines, MCP servers, AI agent workflows, Claude Code, GitHub Copilot, Gemini 2.5 Flash

Cloud & Delivery: AWS (Lambda, API Gateway), GCP, OpenStack, Docker, Kubernetes, CI/CD (GitHub Actions), Jest, automated testing, OAuth 2.0/JWT

Product Domains: Multi-tenant SaaS, HRMS, CRM

Leadership: Technical leadership, mentoring and onboarding, architecture and code review, technical direction, architecture decision records (ADRs)

PROFESSIONAL EXPERIENCE

Backend Lead Engineer

Jan 2024 – Present

MicroAgility Services (Pvt) Ltd. · *Islamabad, Pakistan*

- Built the AI gateway for Talentnix, a multi-tenant SaaS platform: one routing layer in front of OpenAI, Gemini, and Anthropic with automatic failover, which cut AI integration complexity by 60%.
- Designed Talentnix's three-tier resilience pattern (Retry → RAG Fallback → Rule-Based Floor). Users have seen zero AI failures during live LLM outages; the design is written up in the "Designing for AI Failure" case study.
- Enforced tenant isolation across Talentnix's REST and GraphQL APIs with OAuth 2.0/JWT, with no cross-tenant data exposure incidents to date.
- Led backend development for the APAC Management System, MicroAgility's HRMS that combines payroll, HR, and attendance with complete employee records, and for client projects.
- Brought dashboard response times down from 12s to under 2s (83%) through PostgreSQL and MongoDB query tuning, new indexes, and a MeiliSearch rollout.
- Built an event-driven BullMQ and Redis job pipeline that keeps APIs under a second while thousands of async jobs run concurrently.
- Delivered full Jest unit-test coverage with GitHub Copilot, and built an internal ATS with Claude Code and Gemini 2.5 Flash.

- Set up GitHub Actions CI/CD pipelines with automated test gates, shortening release cycles.
- Mentored three associate backend developers through the company's Learning & Development program.

Backend Software Engineer
Mar 2022 – Jan 2024

MicroAgility Services (Pvt) Ltd. · *Islamabad, Pakistan*

- Owned backend development for iAgility and AgileBrains, two large products built around the hiring and consultancy workflow, on a MEAN (MongoDB, Express.js, Angular, and Node.js) microservices architecture.
- Built Express.js microservices and GraphQL APIs that handle each product's hiring flow, with MongoDB as the main data store.
- Wired up Postal mail for transactional email and Google Cloud Storage for files, with WebSocket for real-time notifications.
- Ran zero-downtime schema migrations on live multi-tenant production databases while maintaining 99.9% uptime.
- Refactored the REST and GraphQL services that serve thousands of users across multiple tenants, improving data retrieval speed by about 40%.
- Automated the deployment and migration pipeline for new enterprise tenants, cutting manual onboarding effort.

Software Engineer
Aug 2018 – Mar 2022

Market Icon · *Rawalpindi, Pakistan*

- Worked as the primary full-stack developer on a large-scale MEAN (MongoDB, Express.js, Angular, and Node.js) system, with ownership of backend design and delivery.
- Designed the distributed server and database architecture on an OpenStack cloud platform.
- Modelled the databases for continuously growing real-time data, so reads and writes stayed fast as volume rose.
- Built serverless AWS Lambda and API Gateway services on an event-driven design, removing the idle cost of always-on servers.
- Built Express.js, FastAPI, and Django REST Framework APIs for secure CRM and payment integrations.
- Hardened backend services against peak load and power outages, and introduced code documentation and change tracking.

EDUCATION

B.S. Computer Science · *Government College University · Faisalabad, Pakistan* *Aug 2018*

CERTIFICATIONS

Claude Code in Action (Anthropic) | Containers & Kubernetes Essentials (IBM) | Gen AI: Beyond the Chatbot (Google) | Introduction to Generative AI and Agents (Microsoft) | Certified Django Developer. Verified list in my **portfolio**.

ADDITIONAL INFORMATION

Languages: English (professional working proficiency), Urdu (native)

Availability: Full-time; open to remote, hybrid, or on-site roles, and to relocation with support